Supervised Learning for Gen Z Mental Health Score Prediction from Social Media Usage

Pankaj Kumawat
MSc Data Science
Univ. of Europe for Applied Sciences
Potsdam 14469, Germany
pankaj.kumawat@ue-germany.de

Submitted to
Raja Hashim Ali
Department of Business
Univ. of Europe for Applied Sciences
Potsdam 14469, Germany
hashim.ali@ue-germany.de

***Abstract*—This paper employs supervised regression to predict Gen Z mental health score from the social media usage, platform behavior, demographics and pre-sleep screen time. The dataset includes 1,000,000 records with 12 attributes. The experiments are performed by a reproducible sample of 30,000 records. Seven research questions: are assessed through baseline testing, full model comparison, preprocessing analysis, feature importance, stability of ranking, robustness testing, practical model choices. ElasticNet Regression got the best pure metric result with MAE of 0.777, RMSE of 0.971, R2 of 0.582. The choice was linear regression. final suggested model, as it gave nearly the same accuracy and maximum weighted decision score of 98.248. The the most important predictor was daily hours of use, then addiction level features Simple linear models performed well in general. sufficient for this task, while complex ensemble models did not. That gives you a tangible, practical advantage.**

***Index Terms*—supervised learning, social media analytics, mental health score, regression, Gen Z, feature importance, model robustness, digital wellbeing, scikit-learn**

I. Introduction

Social media is now a normal part of daily life for Gen Z. Users spend time on platforms for communication, entertainment, education, news, and content creation. These usage patterns create structured data that can be studied with supervised machine learning. In this project, the goal is to predict `mental_health_score` from measurable social media and demographic variables.

Mental health score prediction is not a medical diagnosis task in this report. It is treated as a data science problem where behavioral signals are used to estimate a continuous score. The prediction is challenging because wellbeing is affected by many personal and external factors that are not fully captured in a social media dataset. Still, usage intensity, night usage, platform choice, addiction level, and screen time before sleep can provide useful signals for pattern analysis.

This study addresses the problem through seven research questions. It starts with baseline regression models and then compares a wider set of supervised learning methods. It also examines preprocessing effects, feature importance, ranking stability, robustness under noisy or missing data, and a final weighted decision matrix for practical model recommendation.

The main contributions of this work are the following:

- A complete benchmark of supervised regression models for Gen Z mental health score prediction.
- Evidence that simple linear and regularized models perform as well as or better than more complex tree-based models on this dataset.
- Feature importance analysis showing that daily usage hours is the strongest predictor.
- A practical model selection framework that combines accuracy, reliability, interpretability, runtime, and simplicity.

The rest of this paper is organized as follows. Relevant background and related work are reviewed in Section II. The dataset is described in Section III. Section IV defines the regression problem, and Section V presents the research questions. The methodology is explained in Section VI, and the experimental setup is given in Section VII. Results and discussion are reported in Section VIII. Finally, Section IX concludes the paper.

II. Related Work and Background

A. Social Media and Digital Wellbeing

Social media use has been widely studied in relation to adolescent and young adult wellbeing. Existing studies show that social media can be linked with stress, anxiety, sleep problems, social comparison, and overall mental health, but the relationship is not simple or universal [1]. The effect often depends on the user, the platform, the purpose of use, and the surrounding lifestyle.

Recent reviews also warn that digital media variables should not be interpreted as direct clinical evidence [2]. This is important for the present report because the model predicts a score from structured behavior data, not a diagnosis. Therefore, the aim is to understand patterns and compare models, not to make medical claims.

B. Machine Learning for Tabular Prediction

Supervised regression is suitable when the target variable is continuous and labelled examples are available. Common approaches for tabular regression include Linear Regression, Ridge Regression, ElasticNet, Decision Tree, Random Forest, and Gradient Boosting. These models are widely available through the scikit-learn library, which supports preprocessing pipelines, cross-validation, and standard evaluation metrics [3].

Linear models are useful because they are fast and easy to explain. Regularized models such as Ridge Regression and

ElasticNet can improve stability when features overlap or when small coefficient penalties are helpful [4]. Tree-based methods and ensemble models can capture nonlinear relationships, but they may not always improve results when the main signal is close to linear.

C. What Predicts Mental Health Score

In this dataset, the expected predictors include social media usage intensity, platform behavior, night usage, addiction level, and sleep-related screen time. Usage intensity is important because daily exposure may reflect how much social media is part of the user's routine. Addiction level is also relevant because it summarizes a stronger attachment to platform use.

Demographic and platform variables may still be useful, but their effect can be weaker or context-dependent. For that reason, this report uses feature importance analysis instead of assuming which variables matter most. The final interpretation is based on the trained models and generated outputs.

III. Dataset Description

A. Source and Size

The Gen Z Social Media Usage Dataset is stored locally as `data/genz_social_media_usage_1Million.csv`. It contains 1,000,000 rows and 12 columns. Each row describes one user-level record with demographic, usage, platform, and wellbeing-related values. Key facts are summarized in Table I.

TABLE I
DATASET AT A GLANCE

Property	Value
Name	Gen Z Social Media Usage Dataset
Source	Local course CSV dataset
Rows	1,000,000
Columns	12
Target	`mental_health_score`
Task	Supervised regression
Domain	Digital wellbeing and social media analytics
Rows used for modelling	30,000

B. Feature Groups

The dataset contains 11 input features after excluding the target. The features can be organized into four main groups. Demographic features include `age`, `gender`, and `country`. Usage intensity features include `daily_usage_hours`, `avg_session_minutes`, and `num_platforms_used`.

Platform behavior features include `primary_platform`, `purpose`, `night_usage`, and `addiction_level`. Sleep-related behavior is represented by `screen_time_before_sleep`. These variables are suitable for regression because they provide a mix of numeric and categorical information about user behavior.

C. Target Variable

The target variable is `mental_health_score`. It is treated as a continuous value, so the task is regression rather than classification. The goal is to learn how the input variables map to the target score and then evaluate how well different supervised models perform.

IV. Problem Statement

The task is formulated as a supervised regression problem. Each sample consists of a feature vector $\mathbf{x} \in \mathbb{R}^d$ created from one social media usage record. The objective is to learn a function $\hat{f}$ that predicts the mental health score:

$$\hat{y} = \hat{f}(\mathbf{x}) \approx y \tag{1}$$

where y is the observed `mental_health_score`. The model is trained on a labelled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$ and evaluated using a held-out test split and cross-validation. Performance is measured using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

V. Research Questions

This study is guided by seven research questions covering model performance, feature behavior, robustness, and practical deployment.

- **RQ1** How effectively can baseline supervised learning models predict mental health score?
- **RQ2** Which supervised learning model achieves the best predictive performance?
- **RQ3** How do preprocessing strategies affect model performance?
- **RQ4** Which input features contribute most to prediction?
- **RQ5** How does model ranking change across MAE, RMSE, and R^2?
- **RQ6** How robust is the best model under cross-validation, noise, and missing-data conditions?
- **RQ7** Which model provides the best practical balance of accuracy, interpretability, reliability, runtime, and deployment suitability?

VI. Methodology

The overall workflow is illustrated in Fig. 1. It covers data loading, sampling, preprocessing, model training, evaluation, feature analysis, robustness testing, and final model recommendation.

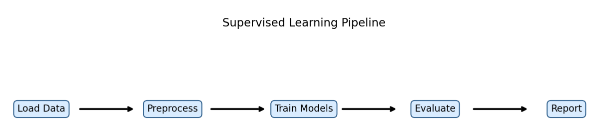

Fig. 1. End-to-end supervised learning pipeline used in this study, covering dataset loading, preprocessing, model training, metric evaluation, robustness testing, and practical model selection.

A. Data Loading and Sampling

The dataset was loaded from CSV format using Python. A reproducible random sample of 30,000 records was used for modelling so that the experiments could run quickly and produce the same outputs each time. A fixed random seed of 42 was used across sampling, splitting, and model training.

The target column was separated from the input features. Numeric and categorical columns were processed differently because the dataset contains both continuous values and text labels. This avoids forcing categorical variables into an incorrect numeric format.

B. Preprocessing Strategies

Three preprocessing strategies were compared in RQ3:

1) **Numeric Only + Scale**: Numeric features were imputed and scaled.
2) **Impute + OneHot + Scale**: Numeric features were imputed and scaled, while categorical features were imputed and one-hot encoded.
3) **Impute + OneHot, No Scale**: Categorical features were encoded, but numeric scaling was not applied.

All preprocessing was implemented through scikit-learn pipeline objects. This helps avoid data leakage because transformations are fitted only on the training data before being applied to test data.

C. Models

The project evaluated seven regression models. These models cover simple baselines, linear models, regularized models, and tree-based methods. Table II summarizes the models used in the experiments.

TABLE II
MODELS USED IN THE STUDY

Model	Role in the Study
Dummy Mean	Mean prediction baseline
Linear Regression	Simple interpretable regression model
Ridge Regression	Regularized linear model
ElasticNet	Regularized linear model with mixed penalty
Decision Tree	Simple nonlinear tree model
Random Forest	Bagging ensemble of decision trees
Gradient Boosting	Boosting ensemble model

D. Evaluation Metrics

All models were evaluated using MAE, RMSE, and R^2.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{2}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} \tag{3}$$

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \tag{4}$$

MAE gives the average absolute prediction error. RMSE gives more penalty to larger errors. R^2 measures how much variance in the target is explained by the model.

E. Feature Importance Analysis

Feature importance was used to identify which inputs contributed most to prediction. The main output ranked the transformed model features by importance. This helps explain the model beyond just reporting accuracy.

F. Robustness Testing

The best-performing model was tested under different conditions. The clean holdout result was compared with 5-fold cross-validation, 15% numeric noise, and 15% missing values. This simulates situations where real-world data may be noisy or incomplete.

G. Multi-Criteria Decision Framework

For the final recommendation, models were scored using accuracy, reliability, interpretability, runtime, and simplicity. These criteria were combined into a weighted decision score. This makes the final model choice more practical than selecting only the best metric value.

VII. EXPERIMENTAL SETUP

All experiments were implemented in Python using scikit-learn, pandas, NumPy, matplotlib, and seaborn. The random seed was fixed at 42 to make the results reproducible. The code generated all tables and figures saved in the project folders.

TABLE III
SOFTWARE CONFIGURATION

Component	Detail
Language	Python 3.x
ML framework	scikit-learn [3]
Data handling	pandas, NumPy
Visualization	matplotlib, seaborn
Random seed	42
Sample size	30,000 rows
Validation	Holdout and 5-fold CV

VIII. RESULTS AND DISCUSSION

A. RQ1 – Performance of the Baseline

The first experiment tested basic supervised models against a simple mean baseline. Table IV shows that Ridge Regression and Linear Regression performed almost the same, with MAE of 0.777, RMSE of 0.972, and R^2 of 0.582. Decision Tree performed slightly worse, while Dummy Mean performed much worse.

The result shows that the input features contain real predictive signal. A simple mean predictor cannot explain the variation in mental health score. The linear models already achieve strong performance, which suggests that the main signal is easy to capture with simple regression.

TABLE IV
RQ1 – BASELINE HOLD-OUT PERFORMANCE

Model	MAE	RMSE	R^2	Train Time
Ridge Regression	0.777	0.972	0.582	0.204
Linear Regression	0.777	0.972	0.582	0.249
Decision Tree	0.797	0.998	0.559	0.421
Dummy Mean	1.217	1.503	0.000	0.236

Fig. 2. RQ1: Error metric comparison for the baseline models. Linear and Ridge Regression show lower error than Decision Tree and much lower error than Dummy Mean.

B. RQ2 – Best Performing Model

RQ2 compared all supervised models. ElasticNet achieved the best pure metric result, with MAE of 0.777, RMSE of 0.971, and R^2 of 0.582. Ridge Regression and Linear Regression were nearly identical. Gradient Boosting, Random Forest, and Decision Tree did not improve over the top linear models.

TABLE V
RQ2 – FULL MODEL COMPARISON ON HOLD-OUT

Model	MAE	RMSE	R^2	Time
ElasticNet	0.777	0.971	0.582	0.100
Ridge Regression	0.777	0.972	0.582	0.097
Linear Regression	0.777	0.972	0.582	0.108
Gradient Boosting	0.779	0.973	0.581	6.848
Random Forest	0.786	0.981	0.574	1.857
Decision Tree	0.797	0.998	0.559	0.219
Dummy Mean	1.217	1.503	0.000	0.095

The main finding is that model complexity did not add much value. Gradient Boosting was close in accuracy but slower. Random Forest and Decision Tree were weaker. This means the final choice should consider interpretability and practical use, not only tiny metric differences.

C. RQ3 – Sensitivity to Preprocessing

RQ3 tested whether preprocessing choices changed the result. Table VI shows that all three preprocessing strategies produced almost identical rounded performance. Numeric Only + Scale had RMSE of 0.971, while the two one-hot encoded pipelines had RMSE of 0.972.

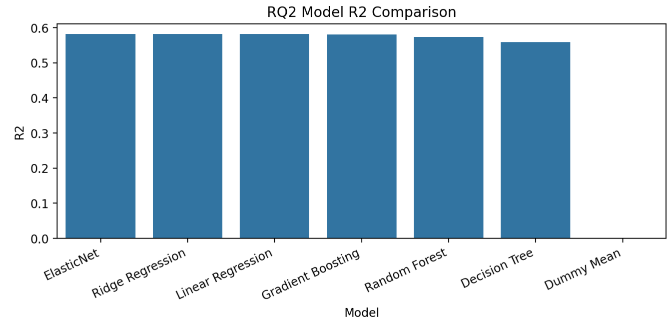

Fig. 3. RQ2: R^2 comparison for all models. ElasticNet, Ridge Regression, and Linear Regression form the strongest group, while Dummy Mean explains no useful variance.

TABLE VI
RQ3 – EFFECT OF PREPROCESSING STRATEGIES

Preprocessing Strategy	MAE	RMSE	R^2
Numeric Only + Scale	0.777	0.971	0.582
Impute + OneHot + Scale	0.777	0.972	0.582
Impute + OneHot, No Scale	0.777	0.972	0.582

This shows that the model result is stable across preprocessing choices. Adding categorical variables through one-hot encoding did not visibly improve the rounded result. The model mainly depends on the strongest numeric and encoded usage features.

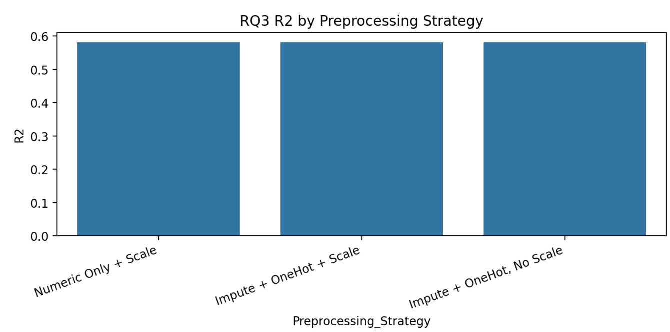

Fig. 4. RQ3: Comparison of R^2 scores across preprocessing strategies. The differences are very small, showing that the result is not dependent on one narrow preprocessing setup.

D. RQ4 – Feature Significance

RQ4 analyzed which variables were most important. The strongest feature was `daily_usage_hours`, with importance value 1.089. This value was much higher than the other predictors, showing that daily usage intensity is the main driver in the model output.

The next strongest predictors were addiction level features. This makes sense because addiction level summarizes the user's relationship with social media usage. Platform, country, number of platforms, and night usage had smaller rounded importance values in this experiment.

TABLE VII
RQ4 – TOP FEATURE IMPORTANCES

Feature	Importance
daily_usage_hours	1.089
addiction_level_High	0.068
addiction_level_Low	0.017
primary_platform_YouTube	0.005
country_India	0.004
num_platforms_used	0.002
night_usage	0.001

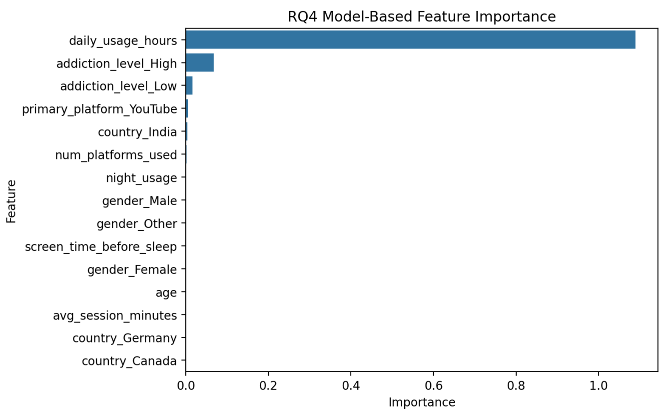

Fig. 5. RQ4: Feature importance values for the trained model output. Daily usage hours dominates the ranking, followed by addiction level features.

E. RQ5 – Stability of the Ranking Across Metrics

RQ5 checked whether the model ranking changed across MAE, RMSE, and R^2. Table VIII shows that ElasticNet ranked first across all three metrics. Ridge Regression and Linear Regression were tied closely, with average rank 1.33.

TABLE VIII
RQ5 – CROSS-METRIC RANKING STABILITY

Model	MAE Rank	RMSE Rank	R^2 Rank	Avg. Rank
ElasticNet	1	1	1	1.00
Ridge Regression	1	2	1	1.33
Linear Regression	1	2	1	1.33
Gradient Boosting	4	4	4	4.00
Random Forest	5	5	5	5.00
Decision Tree	6	6	6	6.00
Dummy Mean	7	7	7	7.00

The ranking is mostly stable. The same top group appears across the metrics, and the same weaker models stay at the bottom. This supports the idea that final selection can focus on practical differences between the top models.

F. RQ6 – Robustness to Noisy and Missing Data

RQ6 tested the best model under different data conditions. The clean holdout result had RMSE of 0.971 and R^2 of 0.582. The 5-fold cross-validation result was close, with RMSE of 0.976 and R^2 of 0.575.

The model was stable under cross-validation and numeric noise. Missing values caused the largest drop, reducing R^2 from 0.582 to 0.498. This shows that data completeness is important for this task.

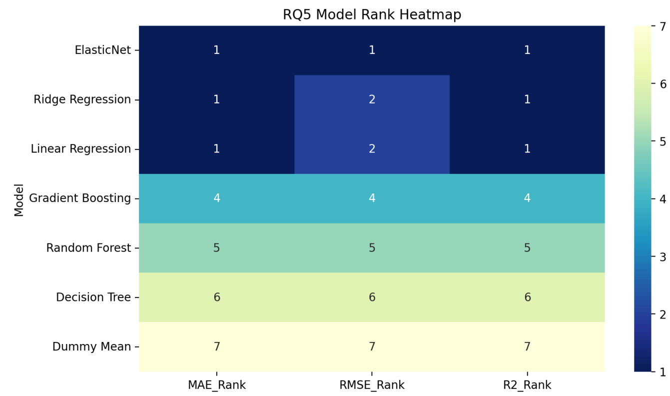

Fig. 6. RQ5: Heatmap of model rankings across MAE, RMSE, and R^2. ElasticNet ranks first, while Ridge Regression and Linear Regression remain very close.

TABLE IX
RQ6 – ROBUSTNESS RESULTS

Scenario	RMSE	R^2	R^2 Drop
Clean holdout	0.971	0.582	0.000
5-fold cross-validation	0.976	0.575	0.007
15% numeric noise	0.984	0.572	0.010
15% missing values	1.065	0.498	0.084

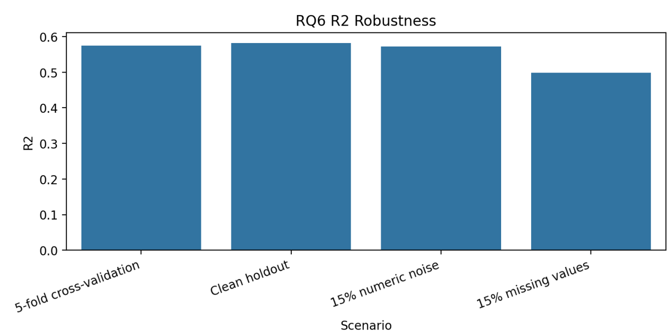

Fig. 7. RQ6: Robustness comparison across clean holdout, cross-validation, noisy input, and missing-value scenarios. Missing data causes the largest performance decrease.

G. RQ7 – Practical Recommendation of Models

RQ7 used a weighted decision matrix to select the most practical model. Linear Regression had the highest final score of 98.248. Ridge Regression scored 96.794, and ElasticNet scored 96.309.

Linear Regression was selected because it is accurate, interpretable, simple, fast, and easy to deploy. ElasticNet is the best pure metric model, but its advantage is very small. For a practical student project and explainable workflow, Linear Regression is the better final recommendation.

TABLE X
RQ7 – WEIGHTED DECISION MATRIX

Model	Acc.	Rel.	Int.	Run.	Simple	Score
Linear Regression	99.812	99.300	95	99.694	95	98.248
Ridge Regression	99.812	99.300	90	100.000	90	96.794
ElasticNet	100.000	99.300	90	99.661	85	96.309
Decision Tree	94.925	98.400	70	98.527	70	88.683
Random Forest	98.120	99.900	45	79.408	45	79.733
Gradient Boosting	99.624	99.400	40	0.000	45	67.248
Dummy Mean	0.000	42.500	100	99.783	100	53.467

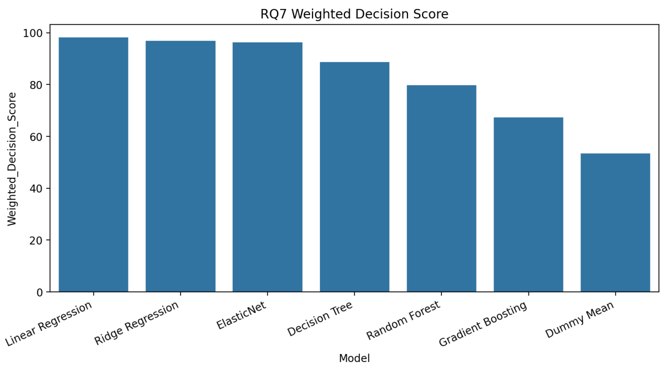

Fig. 8. RQ7: Weighted decision score for practical model selection. Linear Regression achieves the highest overall score.

IX. CONCLUSION

This study investigated supervised learning models for predicting Gen Z mental health score from social media usage data. The dataset contained 1,000,000 records and 12 columns, and the experiments used a reproducible 30,000-row modelling sample. The tested models included Dummy Mean, Linear Regression, Ridge Regression, ElasticNet, Decision Tree, Random Forest, and Gradient Boosting.

The results showed that ElasticNet had the best pure metric performance, with MAE of 0.777, RMSE of 0.971, and R^2 of 0.582. However, Linear Regression achieved almost the same predictive performance and received the highest weighted decision score. This made it the final recommended model for the project.

Feature importance showed that `daily_usage_hours` was the strongest predictor, followed by addiction level features. Preprocessing choices had very little impact on rounded performance, and model rankings were stable across MAE, RMSE, and R^2. Robustness testing showed that the model handled cross-validation and numeric noise well, but missing values caused a larger performance drop.

Future work could test the model on a separate external dataset, add more behavioral variables, and evaluate fairness across demographic groups. More advanced explanation methods such as SHAP could also be used to make the predictions easier to understand.

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